Module – 6

Assignment Submissions – embedUR Networking Course

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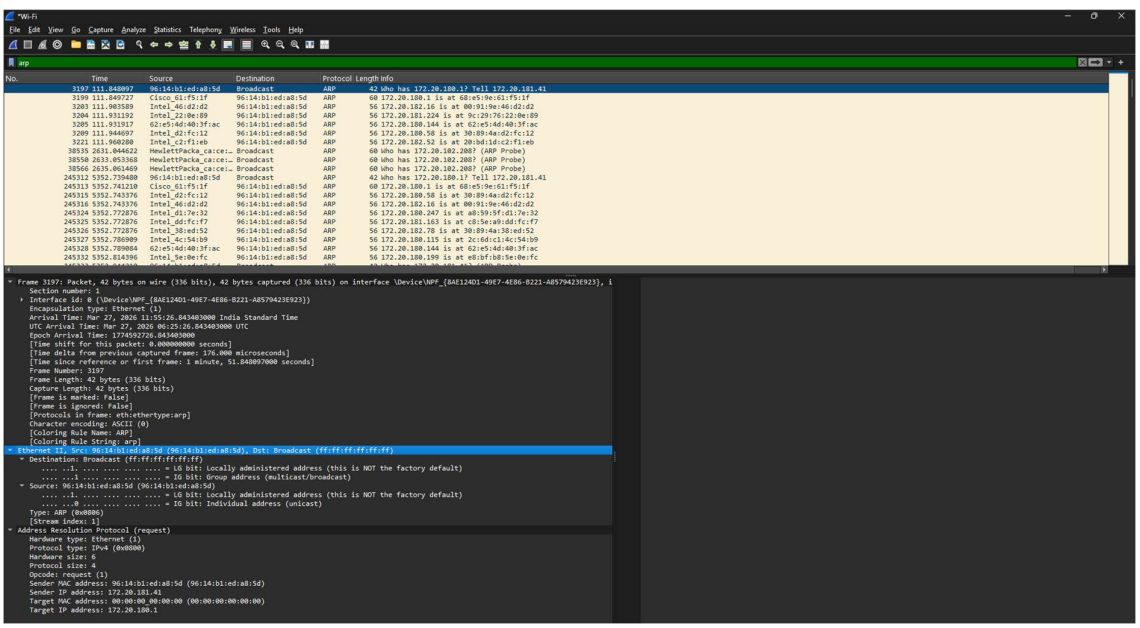
Baalamurugan S P

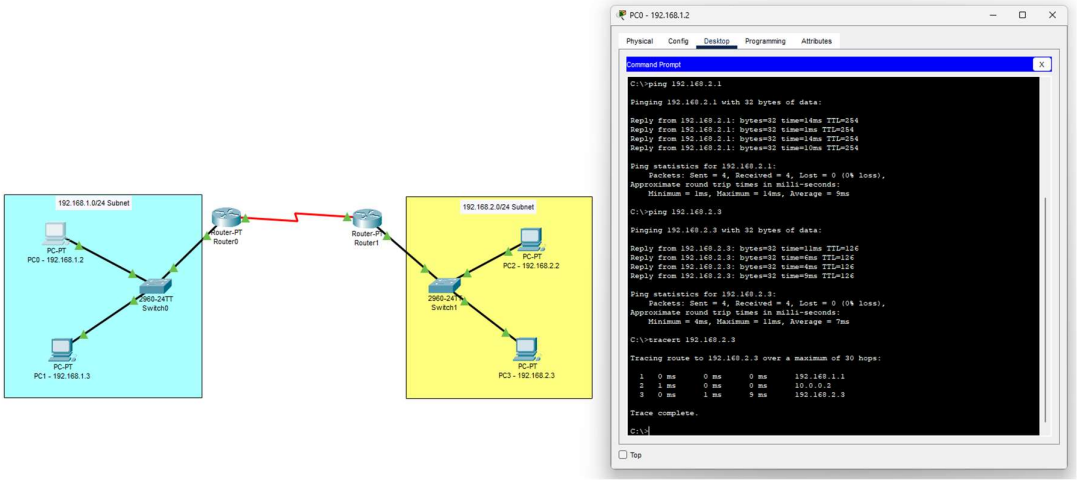
baalamurugan.sp@gmail.com

Q1) **Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address. Discuss the importance of ARP in packet forwarding.**

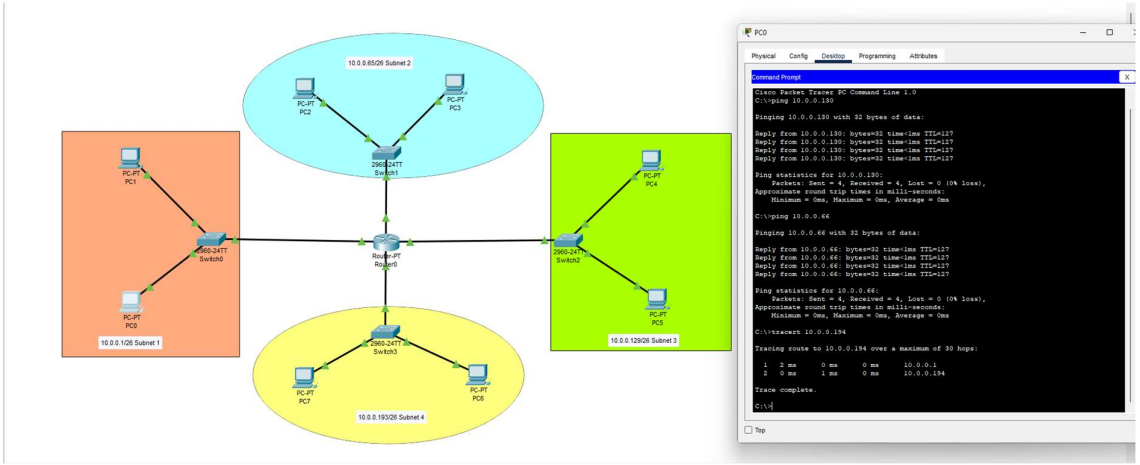


ARP is essential for packet forwarding in a local network because it helps a device find the MAC address of the destination using its IP address. When a device wants to send data, it only knows the IP, but actual transmission happens using MAC addresses. ARP resolves this by mapping the IP to the correct MAC, so the packet can be delivered to the right device. Without ARP, devices wouldn't know where to send data within the network, and communication would fail.

Q2) **Manually configure static routes on a router to direct packets to different subnets. Use the ip route command and verify connectivity using ping and traceroute.**



Q3) **Given a network address of 10.0.0.0/24, divide it into 4 equal subnets. Calculate the new subnet mask. Determine the valid host range for each subnet. Assign IP addresses to devices in Packet Tracer and verify connectivity.**



The network 10.0.0.0/24 can be divided into 4 equal subnets by borrowing 2 bits, resulting in a new subnet mask of 255.255.255.192 (/26).

The four subnets are

10.0.0.0/26 (hosts 10.0.0.1–10.0.0.62),

10.0.0.64/26 (10.0.0.65–10.0.0.126),

10.0.0.128/26 (10.0.0.129–10.0.0.190),

10.0.0.192/26 (10.0.0.193–10.0.0.254),

Each with 64 IP addresses including network and broadcast.

Devices can be assigned IPs from these ranges and connectivity can be verified using ping.

Q4) **You are given three IP addresses: 192.168.10.5, 172.20.15.1, and 8.8.8.8. Identify the class of each IP address. Determine if it is private or public. Explain how NAT would handle a private IP when accessing the internet.**

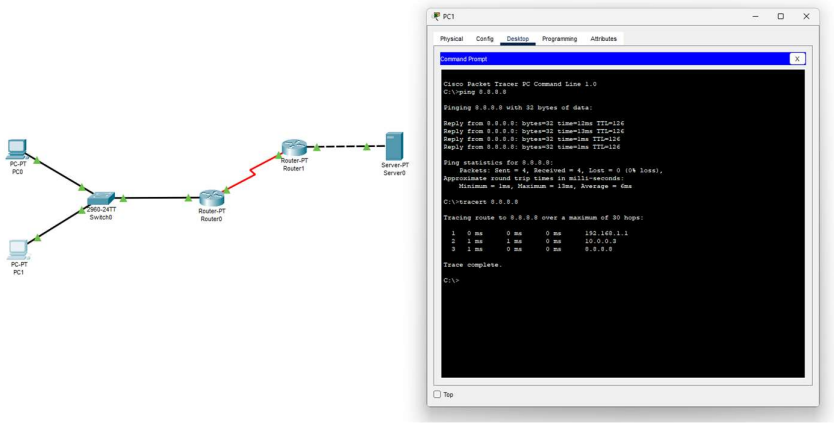
192.168.10.5 – is Class C Default subnet is 255.255.255.0 Range – 192.0.0.0 – 223.255.255.255 Private IP – falls in the 192.168.0.0 – 192.168.255.255 private range

172.20.15.1 – is Class B Default subnet is 255.255.0.0 Range – 128.0.0.0 – 191.255.255.255 Private IP – falls in the 172.16.0.0 – 172.31.255.255 private range

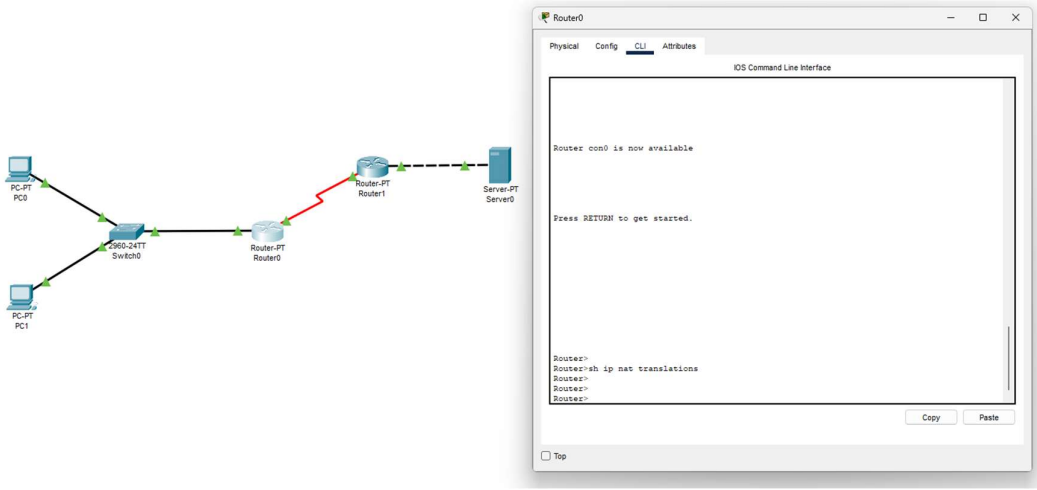
8.8.8.8 – is Class A Default subnet is 255.0.0.0 Range – 1.0.0.0 – 126.255.255.255 Public IP – accessible over the internet (Google DNS)

When a private IP like 192.168.10.5 accesses the internet, NAT on the router translates it to a public IP before the packet leaves the network. The destination server only sees the public IP. When the reply comes back, NAT translates it back to the private IP and forwards it to the correct device.

Q5) **In Cisco Packet Tracer, configure NAT on a router to allow internal devices (192.168.1.x) to access the internet. Test connectivity by pinging an external public IP. Capture the traffic in Wireshark and analyze the source IP before and after NAT translation.**



Before NAT




```
Router(config)#ip nat inside source static 192.168.1.2 10.0.0.2

Router(config)#

Router(config)#interface FastEthernet0/0

Router(config-if)#ip nat inside

Router(config-if)#ex

Router(config)#

Router(config)#interface Serial2/0

Router(config-if)#ip nat outside

Router(config-if)#ex

Router(config)#

Router(config)#do wr

Building configuration...

[OK]
```

```
Router#
Router#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  10.0.0.2          192.168.1.2        ---                ---
```

After NAT and Pinging from PC0 to simulated server (8.8.8.8)

```
Router#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.2:18       192.168.1.2:18     8.8.8.8:18         8.8.8.8:18
icmp 10.0.0.2:19       192.168.1.2:19     8.8.8.8:19         8.8.8.8:19
icmp 10.0.0.2:20       192.168.1.2:20     8.8.8.8:20         8.8.8.8:20
icmp 10.0.0.2:21       192.168.1.2:21     8.8.8.8:21         8.8.8.8:21
---  10.0.0.2          192.168.1.2        ---                ---
```

Before NAT, the source IP address is 192.168.1.2, which is the private IP of the PC inside the local network. After NAT, the source IP address is translated to 10.0.0.2, which is the public (inside global) IP address used by the router to communicate with external networks.